

SymPy Bug Card

Bug 11: `function_range` gives `EmptySet` for $\sqrt{x^2}$

Task. Compute the real range of the square root of a square.

$$\text{function_range}\left(\sqrt{x^2}, x, \mathbb{R}\right)$$

SymPy's answer.

$$\text{function_range}\left(\sqrt{x^2}, x, \mathbb{R}\right) = \emptyset$$

Correct answer.

$$[0, \infty)$$

Explanation. For real x , $\sqrt{x^2} = |x|$, whose range is $[0, \infty)$, not $\emptyset$.

Likely root cause. The diagnosis identifies `function_range` in `sympy/calculus/util.py`: it samples endpoint limits and solutions of $f'(x) = 0$, but misses nondifferentiable interior extrema such as the cusp of $\sqrt{x^2}$ at $x = 0$. Deduplication classifies this as a specific new instance of the known `function_range` / `Abs`-cusp failure family already reported in SymPy issues [#16469](#), [#21956](#), and [#20101](#).

Artifact (GitHub).

[bug-report/bug-011-function-range-gives-emptyset-for-sqrt-x-2/](#)